VENDORGATE

Vendor Onboarding and Purchase Order Portal

REQUIREMENT TRACEABILITY MATRIX (RTM)

Case Study No. 88

1. Purpose

The Requirements Traceability Matrix (RTM) establishes a clear link between VendorGate requirements, system design elements and verification activities. It ensures that each Functional Requirement (FR) is represented in the design and has at least one corresponding test case, as required by the case study.

2. Traceability Approach

The RTM follows the chain: Functional Requirement → Use Case / Design Component → Test Case → Test Evidence / Status. Requirement identifiers follow the VendorGate SRS. Test case identifiers follow the VendorGate Test Plan & Evidence document.

Test execution status is intentionally shown as 'Pending Execution' where actual prototype execution evidence has not yet been captured. This avoids inventing test results.

3. Functional Requirement Traceability

FR ID	Functional Requirement	Related Design / Use Case	Test Case	Test Status	Traceability
FR-01	Vendor registration with required vendor information	Register Vendor / Vendor Registration	TC-01	Pending Execution	Traceable
FR-02	Validate mandatory vendor information before submission	Vendor Registration / Validation	TC-02	Pending Execution	Traceable
FR-03	Manage and update vendor information	Vendor / Vendor Registration	TC-03	Pending Execution	Traceable
FR-04	Create a Purchase Order with required details	Create Purchase Order / PurchaseOrder	TC-04	Pending Execution	Traceable
FR-05	Submit a Purchase Order for approval	Submit Purchase Order for Approval / PO Workflow	TC-05	Pending Execution	Traceable
FR-06	Assign Level 1 approval for PO value up to ₹50,000	Approval-Level Decision Flow / ApprovalPolicy	TC-06	Pending Execution	Traceable

FR-07	Assign Level 2 approval for PO value above ₹50,000 up to ₹2,00,000	Approval-Level Decision Flow / ApprovalPolicy	TC-07	Pending Execution	Traceable
FR-08	Assign Level 3 approval for PO value above ₹2,00,000 up to ₹10,00,000	Approval-Level Decision Flow / ApprovalPolicy	TC-08	Pending Execution	Traceable
FR-09	Assign Level 4 approval for PO value above ₹10,00,000	Approval-Level Decision Flow / ApprovalPolicy	TC-09	Pending Execution	Traceable
FR-10	Allow approver to approve or reject a Purchase Order	Approve / Reject Purchase Order / Approval	TC-10	Pending Execution	Traceable
FR-11	Allow vendor to acknowledge an approved Purchase Order	Acknowledge Purchase Order / VendorAcknowledgement	TC-11	Pending Execution	Traceable
FR-12	Allow review of approval/audit records	Review / Audit Approval Records / Approval	TC-12	Pending Execution	Traceable

4. Business Requirement to Functional Requirement Mapping

The table below links the major business needs represented in the case study to the functional requirement groups. This is a planning-level mapping; the case study does not provide its own BR numbering.

Business Need / Case-Study Area	Related FRs	Design Area	Verification
Online vendor registration replacing email-based document exchange	FR-01 to FR-03	Vendor Registration	TC-01 to TC-03
Controlled Purchase Order creation and submission	FR-04 to FR-05	Purchase Order Workflow	TC-04 to TC-05
Approval-limit enforcement	FR-06 to FR-09	Approval Policy / Approval-Level Flow	TC-06 to TC-09
Approval decision and auditability	FR-10 and FR-12	Approval / Audit Records	TC-10 and TC-12
Vendor acknowledgement of approved PO	FR-11	Vendor Acknowledgement	TC-11

5. UML / Design Traceability

Design Artifact	Requirements / Area Covered	Related Test Cases
Use Case Diagram	FR-01 to FR-12 through actors and major system functions	TC-01 to TC-12
Class Diagram	Vendor, User, PurchaseOrder, Approval, ApprovalPolicy, VendorAcknowledgement	TC-01 to TC-12
Sequence Diagram — Raise and Approve PO	FR-04 to FR-11	TC-04 to TC-11
Activity Diagram	End-to-end PO workflow and decision points	TC-04 to TC-11
PO State Diagram	Draft → Submitted → PendingApproval → Approved/Rejected → Acknowledged	TC-04, TC-05, TC-10, TC-11
Approval-Level Decision Flow	FR-06 to FR-09 and approval boundaries	TC-06 to TC-09

6. Test Technique Traceability

Technique	Primary Coverage	Test Cases
Boundary Value Analysis	Approval-limit boundaries	TC-06 to TC-09
Equivalence Class Testing	Approval value classes and vendor validation classes	TC-01 to TC-03, TC-06 to TC-09
Decision Table Testing	Approval-level decision logic	TC-06 to TC-09
Functional Testing	Core VendorGate workflows	TC-01 to TC-12

7. Non-Functional Requirement Traceability

The SRS contains eight measurable NFRs. Because the current Test Plan is primarily functional and prototype-oriented, the following matrix identifies the planned verification method. Final measured values should be populated after execution; no unsupported performance/security figures are invented here.

NFR	Verification Method	Evidence	Status
NFR-01	Usability / UI walkthrough	Screenshots and checklist	Pending Execution
NFR-02	Performance measurement under prototype test conditions	Execution log / timing evidence	Pending Execution
NFR-03	Availability/reliability observation during demo test	Execution log	Pending Execution
NFR-04	Input validation and error-handling checks	Screenshots / test results	Pending Execution
NFR-05	Access/control behaviour review	Test evidence	Pending Execution

NFR-06	Data consistency checks in prototype flows	Test results	Pending Execution
NFR-07	Maintainability/code-structure review	Code review checklist	Pending Execution
NFR-08	Responsive/interface compatibility check	Screenshots across supported viewport sizes	Pending Execution

8. Traceability Completeness Summary

Traceability Check	Result
Total Functional Requirements	12
FRs mapped to at least one test case	12 / 12
FRs mapped to design/use-case elements	12 / 12
Functional test execution completed	Pending prototype execution
Evidence captured for all functional tests	Pending prototype execution
Final defect metrics available	Pending prototype execution

9. RTM Maintenance & Change Control

The RTM should be updated whenever a requirement changes, a design element is added or removed, a test case changes, or a defect causes a requirement to be re-tested. Requirement, test-case and evidence identifiers should remain consistent across the SRS, UML/System Design and Test Plan.

10. Conclusion

The VendorGate RTM provides end-to-end traceability from the 12 Functional Requirements through system design and verification. It completes the planned documentation chain and provides a single reference for checking requirement coverage before the prototype demonstration and final testing.